

FASER V10

Architecture & workflow — data pipeline, build graph, modules

André Rubbia

October 3, 2026

IPA Rubbia Group, ETH Zürich

Data processing pipeline

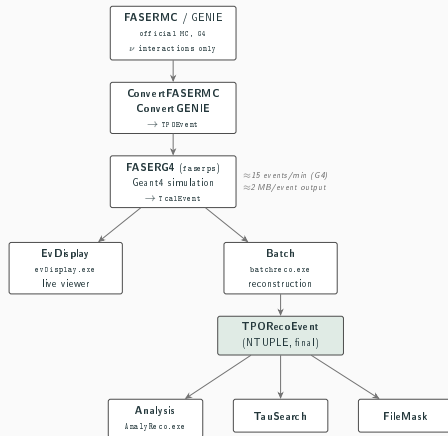
External dependency build graph

Repository & module architecture

Where to look

Data processing pipeline

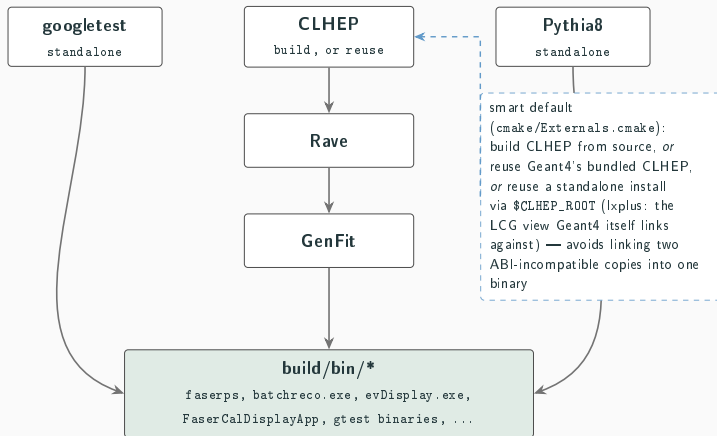
Data processing pipeline



Same chain for every sample type; only the input source (official FASERMC or a GENIE conversion) changes which Convert step runs.*

External dependency build graph

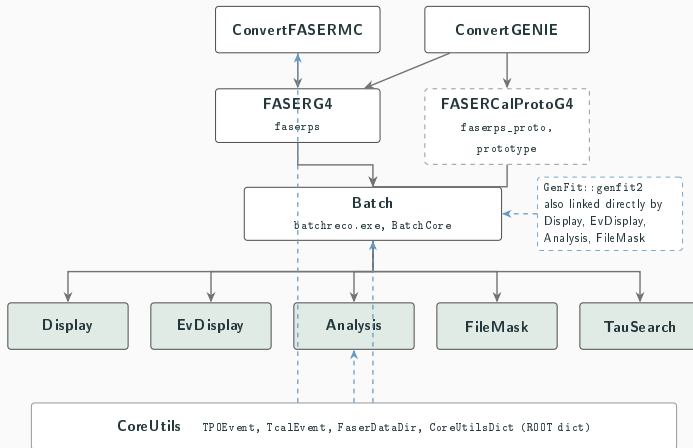
External dependency build graph



FASER_BUILD_EXTERNALS=ON (default) builds all five as a CMake superbuild before configuring FASER's own packages.

Repository & module architecture

Repository & module architecture



Solid arrows: build/data flow (CMakeLists.txt add_subdirectory order). Dashed arrows: shared-library dependency (target_link_libraries). Mint: user-facing leaf executables.

Where to look

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- `cmake/Externals.cmake` — the superbuild in this talk's second diagram: CLHEP/Rave/GenFit/googletest/Pythia8, including the CLHEP reuse smart-default (from-source / Geant4-bundled / `$CLHEP_ROOT`)
- `docs/INSTALL.md` — full CMake build: prerequisites, every `FASER_BUILD_*` option, troubleshooting a clean build
- `docs/GEANT4_INSTALL.md` — building/installing Geant4 itself
- `run_faserps.md` / `run_batchreco.md` — prose docs for the pipeline's two wrapper scripts (first diagram)
- `docs/REGRESSION_TESTS.md` — the regression suite (`run_regression_tests.py`): simulate → reconstruct → compare against a committed golden baseline, end to end